CS211 Review 2 Answers

1. On DoltHub, why would you make changes to a database on a branch instead of directly on `main`?

A) So the data on `main` stays the same until you check your changes and merge them
B) So the database gets new tables automatically
C) So the changes can never be merged
D) So other people can't see the database

Answer: A - Branches work the same way in DoltHub as in GitHub: `main` doesn't change until you merge.

2. In GitHub, you create a branch called `fix-header`, change `index.html` on that branch, and commit and push the change to your repository. What happens to `index.html` on `main`?

A) It has the same change
B) It is deleted
C) It stays the same; the change is only on `fix-header` until you merge it
D) It is renamed to `fix-header`

Answer: C - A commit on a branch stays on that branch. `main` gets the change only when the branch is merged.

3. On DoltHub, you merge your branch into `main` and then delete the branch. What happens to the changes you made on it?

A) They are removed from `main`
B) They stay in `main`
C) They move to a new database
D) They go back onto a new branch

Answer: B - Merging copied the changes into `main`; deleting the branch doesn't undo that. It works the same way in GitHub.

4. You want to publish a website, and you also want to keep a products and orders database that you can query with SQL. Which should you use for each?

A) DoltHub for the website, GitHub for the database
B) GitHub for both
C) DoltHub for both
D) GitHub (with GitHub Pages) for the website, DoltHub for the database

Answer: D - GitHub Pages publishes websites; DoltHub stores databases and has a SQL editor.

5. Which of these do both GitHub and DoltHub have?

A) Branches, commits, and merging into `main`
B) A SQL editor for running queries
C) GitHub Pages websites
D) Codespaces

Answer: A - DoltHub copies Git's ideas: branches, commits, merges, and forks all work the same way.

6. What is an AI agent, like OpenCode, Claude Code, or Codex?

A) A person who is a collaborator on the repository
B) A GitHub setting for private repositories
C) An AI helper that can edit files and run commands for you
D) A type of database table

Answer: C - You describe a task and the agent works directly in your files.

7. Which of these is true about a primary key?

A) Two rows can share the same value
B) Every row has a different value
C) It must be a text column
D) It is always the last column

Answer: B - Primary key values are unique: no two rows share one.

8. In a database, what is a foreign key?

A) A column that refers to the primary key of another table
B) A key from a database in another country
C) A column that must be text
D) A copy of the whole table

Answer: A - A foreign key connects a row in one table to a row in another.

9. Which of these would **not** be a good primary key for a table of people?

A) Email address

B) First and last name
C) Phone number
D) Customer ID number

Answer: B - Email addresses, phone numbers, and ID numbers each belong to one person, but two people can have the same name.

10. Why is a person's first name and last name not a good primary key?

A) Names are too long to store
B) Two different people can have the same name, like two John Smiths
C) Names have to be kept in a separate table
D) A primary key can only be one letter

Answer: B - A primary key must be unique, and names are not. Names can also change.

11. Why would you save a query as a view?

A) So the query's tables are deleted
B) So the database becomes private
C) So anyone can rerun it by name
D) So the query runs only once

Answer: C - A view like `customer_spending` saves the work of writing the query again.

12. On DoltHub, how do you save a query as a view?

A) Upload it as a file named `view.sql`
B) Paste it into the database's README
C) Copy and paste it into Gemini
D) Run it in the SQL editor, then click **Create view** under the result and give the view a name

Answer: D - Name it with lowercase letters and underscores, like `customer_spending`.

13. Which of these is part of a database's schema, not its data?

A) The column `price` is `DECIMAL(10,2)`, and `product_id` is the primary key
B) Jane Smith ordered a Silk Blouse
C) There were 10 orders in March
D) The Leather Jacket costs 249.99

Answer: A - Column names, data types, and keys describe structure; the other choices are rows of data.

14. Before you ask Gemini to write SQL queries for your database, why do you paste the table schemas into the chat first?

A) So Gemini can change your database
B) So Gemini can log in to DoltHub
C) So Gemini knows the table names, column names, and how the tables connect, and writes queries that fit them
D) So the database becomes public

Answer: C - Gemini can't see your database; the schema tells it what is there.

Questions 15 to 20 refer to this small database. Order 6 has not been filled in yet.

Customers

customer_id	first_name	last_name
2	Jane	Smith
3	Robert	Johnson
4	Sarah	Williams

Products

product_id	name	price
3	Cashmere Sweater	149.99
4	Leather Jacket	249.99
5	Cotton T-Shirt	29.99

Orders

order_id	customer_id	product_id	order_date
4	4	3	2025-03-20
5	3	4	2025-03-22
6	?	?	2025-03-25

15. In the Orders table, which column links each order to a row in the Products table?

A) `order_id`
B) `customer_id`
C) `order_date`
D) `product_id`

Answer: D - `product_id` is the foreign key that refers to the Products table's primary key.

16. Which of these statements is **not** true?

A) `product_id` 4 can appear on many rows of the Orders table
B) Robert Johnson, customer 3, can place many orders
C) Two rows of the Products table can both have `product_id` 5
D) Every row of the Orders table has its own `order_id`

Answer: C - `product_id` is the primary key of Products, so no two products share one. In Orders it is a foreign key, so it can repeat.

17. Who placed order 5, and what did they buy?

A) Robert Johnson bought a Leather Jacket
B) Sarah Williams bought a Cashmere Sweater
C) Sarah Williams bought a Leather Jacket
D) Robert Johnson bought a Cotton T-Shirt

Answer: A - Order 5 has `customer_id` 3 (Robert Johnson) and `product_id` 4 (Leather Jacket).

18. What was the price of the product bought in order 4?

A) 249.99
B) 29.99
C) 149.99
D) You can't tell, because the Orders table has no price column

Answer: C - Order 4 has `product_id` 3. Looking up product 3 in Products gives the Cashmere Sweater at 149.99.

19. On 2025-03-25, Jane Smith buys a Cotton T-Shirt, and it is recorded as order 6. What goes in place of the two `?` marks for `customer_id` and `product_id`?

A) `customer_id` 5, `product_id` 2
B) `customer_id` 2, `product_id` 5
C) `customer_id` 2, `product_id` 3
D) `customer_id` Jane Smith, `product_id` Cotton T-Shirt

Answer: B - Jane Smith is customer 2 and the Cotton T-Shirt is product 5.

20. One possible view for these tables would show each order with the customer's last name, the product name, and the product price. Which of these could **not** be made into a view from these tables?

A) Each order with the customer's first and last name
B) Each order with the store where it was bought
C) The number of orders placed on each date
D) Each order's date with the product's price

Answer: B - A view can only use columns that are in the tables. None of these tables records a store.

21. What does SQL stand for?

A) Simple Question Language
B) Stored Query List
C) Structured Query Language
D) System Quality Logic

Answer: C - SQL is the language we use to talk to a relational database.

22. A SQL script creates a `products` table and an `orders` table that has a foreign key to `products`. Which table must be created first?

A) `orders`, because it has more rows
B) Either one, the order does not matter
C) Neither, they are created together automatically
D) `products`, because `orders` can only refer to a table that already exists

Answer: D - A foreign key can only point to a table that is already there.

23. A Products table has the columns `product_id`, `name`, `price`, and `in_stock`. Which column is the primary key?

A) `name`
B) `price`
C) `in_stock`
D) `product_id`

Answer: D - `product_id` gives each product a unique ID.

24. In SQL, what does a `JOIN` do?

A) Deletes rows that two tables share
B) Creates a new database
C) Merges a branch into `main`
D) Matches a foreign key in one table to the primary key in another, so their columns appear together

Answer: D - A join lets one query show, for example, each order with its product name.